

Let B_k be unreserved quote balance before interval k . Assuming zero new inflow, regular 8-hour ticks, full execution, and negligible dust:

$$B_{k+1} = B_k \cdot (1 - \alpha) = B_k \cdot \left(\frac{20}{21}\right)$$

By mathematical induction:

$$B(k) = B_0 \cdot \left(\frac{20}{21}\right)^k$$

With a constant fee inflow of I every 8 hours, steady-state balance satisfies $B_{eq} \cdot \alpha = I$:

$$B_{eq} = \frac{I}{\alpha} = 21 \cdot I$$

Ideal Steady State: Under identical assumptions, steady-state balance is $21I$, equivalent to seven days of continuous inflows. Actual balance varies with batch arrivals, keeper delays, liabilities, and rounding.

2.2 Half-Life Analytical Derivation

Setting $B(k) = \frac{1}{2}B_0$, the half-life interval count $k_{\frac{1}{2}}$ satisfies:

$$\left(\frac{20}{21}\right)^{k_{\frac{1}{2}}} = \frac{1}{2} \Rightarrow k_{\frac{1}{2}} = \frac{\ln(0.5)}{\ln(\frac{20}{21})} \approx 14.207 \text{ intervals}$$

Converting to physical elapsed time $t_{\frac{1}{2}}$:

$$t_{\frac{1}{2}} = k_{\frac{1}{2}} \times 8 \text{ hours} \approx 113.654 \text{ hours} \approx 4.736 \text{ days}$$

Elapsed Time	Interval k	Remaining %	Exact Value
Day 1 (24h)	3	$\left(\frac{20}{21}\right)^3$	86.38%
Day 3 (72h)	9	$\left(\frac{20}{21}\right)^9$	64.46%
Day 7 (168h)	21	$\left(\frac{20}{21}\right)^{21}$	35.89%
Day 14 (336h)	42	$\left(\frac{20}{21}\right)^{42}$	12.88%
Day 30 (720h)	90	$\left(\frac{20}{21}\right)^{90}$	1.24%

Reserve capital is not mathematically exhausted after 7 days of zero volume; after 30 days, approximately 1.24% of the initial balance remains.

2.3 Dynamic Time-Proration and Discrete Rate Limits

1. Single-Release Span Ceiling (CAP_SPAN):

In `IntervalController.sol`, maximum prorated time is capped at $CAP_SPAN = 2 \times MIN_INTERVAL = 16$ hours:

$$c_{i(m)} = \begin{cases} 0 & \text{if } m < \text{activeFrom}_i \\ 1 + \min(m - \text{effectiveFrom}_i, 21) & \text{if } m \geq \text{activeFrom}_i \text{ and } m < \text{endInterval}_i \end{cases}$$

$$\text{Allocation} = \left\lfloor (\text{accounted} - \text{claims}) \cdot \frac{\min(\Delta t, CAP_SPAN)}{T_{\text{week}}} \right\rfloor$$

2. Vault Outflow Rate Limit (windowAllowance):

In `Belly.sol`, immutable parameters restrict maximum window outflow:

$$\text{MaxOutflow} = \left\lfloor \text{Balance} \cdot \frac{16}{168} \right\rfloor \approx 9.5238\%$$

Even under compromised spenders, continuous 50% drawdown equivalent is:

$$n_{\frac{1}{2}} = \frac{\ln(0.5)}{\ln(\frac{152}{168})} \approx 6.92569 \text{ windows} \approx 55.4055 \text{ hours}$$

Discrete execution requires seven full window allowances (49.6295% remaining after 7th). Spanning 6 intervals, it takes 48 hours if initial allowance is usable, and 56 hours if waiting for window rollover.

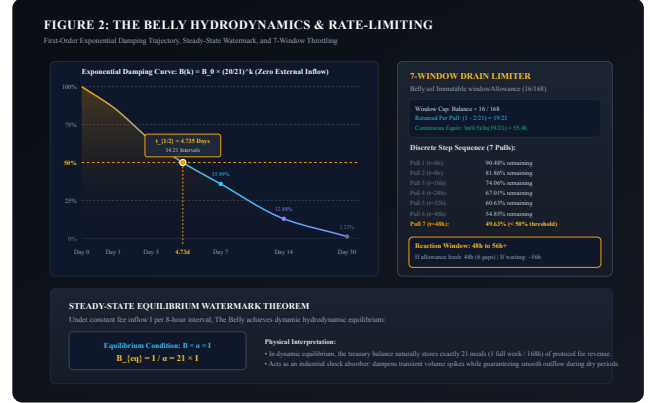


Figure 2: The Belly Hydrodynamic Decay Curve and Discrete Rate-Limiter Defensive Model

3 Seniority Ledger & Scalable Topology

3.1 Independent Positions and State Isolation

Each FATCAT stake creates an isolated position across two contracts:

$\text{Position}_i = \{\text{owner}, \text{principal}, \text{diet}, \text{activeFrom}, \text{effectiveFrom}, \text{endInterval}\}$

- Anti-merging:** Positions cannot be merged, preventing seniority whitewashing;
- Next-interval Activation:** Stakes in interval m become active at $\text{activeFrom} = m + 1$, neutralizing flash-loan dividend exploits;
- Effective Activation Mapping:** For standard positions, $\text{effectiveFrom} = \text{activeFrom}$; for positions opened with a Seniority Certificate (via `stakeWithCertificate`), let certificate head-start be $\text{head} = \text{multiplier} - 1$, setting $\text{effectiveFrom} = \text{activeFrom} - \text{head}$.

3.2 $O(1)$ Closed-Form Seniority Aggregation

Position i with principal p_i accrues seniority notch $c_{i(m)}$. Prior to activation, coefficient is 0; once active, it scales from virtual activation interval effectiveFrom_i up to 22:

Positions divide into:

- Climbing Cohort:** $m - \text{effectiveFrom}_i < 21$;
- Settled Cohort:** $m - \text{effectiveFrom}_i \geq 21$.

Summing over climbing positions:

$$\sum_{i \in \text{Climb}} p_i (1 + m - \text{effectiveFrom}_i) = (1 + m) \sum p_i - \sum (p_i \text{effectiveFrom}_i)$$

The contract maintains four global scalars: $P_{\text{climb}} = \sum_{i \in \text{Climb}} p_i$, $J_{\text{climb}} = \sum_{i \in \text{Climb}} (p_i \cdot \text{effectiveFrom}_i)$, $P_{\text{settled}} = \sum_{i \in \text{Settled}} p_i$, $\text{headOpening} = \sum_{i \in \text{Opening}} (\text{head}_i \cdot p_i)$ (Opening interval head-start weight)

Total active weight evaluation is strictly $O(1)$, subtracting transient headOpening in the opening interval:

$$W(m) = (1 + m)P_{\text{climb}} - J_{\text{climb}} + 22P_{\text{settled}} + W_{\text{exiting}} - \text{headOpening}$$

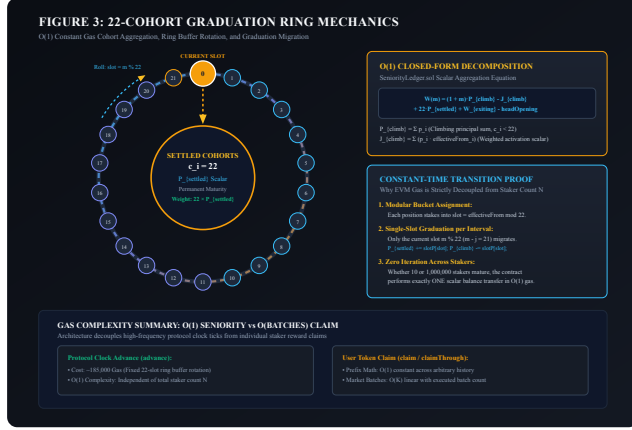


Figure 3: 22-Cohort Graduation Ring Mechanics

3.3 Dual-Accumulator Formulation

Per-interval yield per unit weight for asset a scaled by RAY = 10^{27} :

$$\omega_{a,m} = \left\lfloor \frac{\text{Share}_{a,m} \cdot \text{RAY}}{W_{\text{open}}(a, m)} \right\rfloor$$

Prefix accumulators:

$$A_{a,m} = A_{a,m-1} + (1 + m) \cdot \omega_{a,m}$$

$$B_{a,m} = B_{a,m-1} + \omega_{a,m}$$

Climbing position quote entitlement:

$$\text{Entitlement}(j, m) = \left\lfloor \frac{p \cdot (\Delta A_a - j \cdot \Delta B_a)}{\text{RAY}} \right\rfloor$$

Settled position quote entitlement:

$$\text{Entitlement}_{\text{settled}} = \left\lfloor \frac{22 \cdot p \cdot \Delta B_a}{\text{RAY}} \right\rfloor$$

Dual prefix subtraction yields $O(1)$ entitlement arithmetic per clean interval. Complete token redemption requires $O(K)$ batch iteration, boundable via `claimThrough`.

3.4 Mid-Interval Forfeiture Netting

Positions can redeem principal without lockup penalties (`redeem()`), forfeiting the active interval's meal:

$$\text{Forfeited}_a = \left\lfloor \text{Allocation}_a \cdot \frac{w_{\text{exit}}}{W_{\text{open}}(a)} \right\rfloor$$

Forfeited funds remain permanently in asset a 's pot, reducing denominator for subsequent batch execution: $\text{ClaimablePot}_a = \text{Pot}_a - \text{Forfeited}_a$, increasing `BatchRate` for remaining participants.

4 Allocation Neutrality & Seniority Scenarios

4.1 Nominal Quote Allocation Under Ideal Conditions

Under security property S10, $\sum_{a \in \text{MENU}} W_{\text{open}}(a) = W_{\text{total}}$. Asset a allocation is:

$$Q_a = Q_{\text{total}} \cdot \frac{W_{\text{open}}(a)}{W_{\text{total}}}$$

Position i 's entitlement is:

$$\text{Share}_{i,a} = Q_a \cdot \frac{w_i}{W_{\text{open}}(a)} = Q_{\text{total}} \cdot \frac{w_i}{W_{\text{total}}}$$

Denominators cancel in real numbers without probation caps. Position quote budget is $\frac{w_i}{W_{\text{total}}}$, independent of cohort size.

4.2 Seniority Relative Advantage Analysis

Comparison	Multiplier	Economic Context
Max Mature vs Notch 1	22.0x	Temporary disparity, parity in 21 ticks
Max Mature vs Pool Average	$\frac{22}{\bar{c}}$	Distribution-dependent, $\bar{c} \in [1, 22]$
Mature in Stagnant Pool	1.0x	All mature, premium converges to 1

With principal-weighted average seniority $\bar{c} \in [1, 22]$, relative advantage $\frac{22}{\bar{c}}$ spans 1.0x to 22.0x. Uniform distribution ($\bar{c} = 11.5$) yields 1.913x as a representative scenario, not a mathematical upper bound.

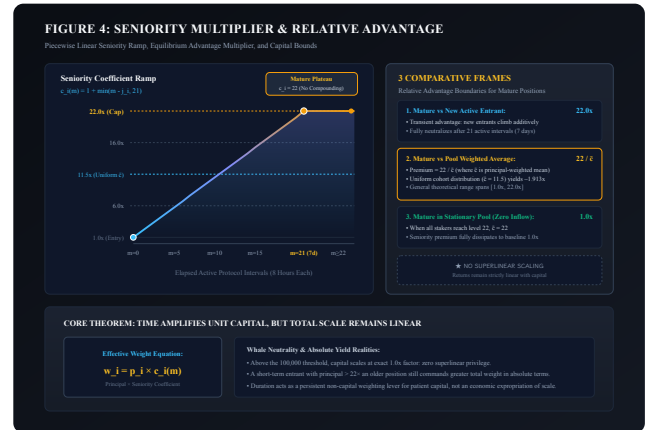


Figure 4: Seniority Multiplier, Settled Premium Curve, and Linear Principal Bounds

4.3 Minimum Stake and Seniority Certificate Specification

- Immutable Minimum Stake:** `minStake` set to 100_000 FATCAT (0.01% supply), no setter post-deployment;
- Strict Linear Scaling:** Weight scales strictly linearly with capital beyond threshold;
- Seniority Physicalization Channel:**

- **Genesis Deployment:** Deployed alongside core staking contracts. Stakers fully exiting standard positions with $\geq 100,000$ FATCAT after 21 days can call `redeemAndIssueCertificate()` to imprint their achieved notch ($c_i \in [1, 22]$) into an ERC-721 credential;
- **Permanent Dead Address Lock:** Transfers 100,000 FATCAT to blackhole `0x...dEaD` (via `transfer(DEAD, MINT_BURN)`), not reducing total supply via `burn()`; remaining principal refunded;
- **Certificate**
Staking: Call `stakeWithCertificate(principal, diet, certificateId)` to inherit starting multiplier, exempt from `minStake`; locks certificate during tenure; no re-minting;
- **Hard Cap & Unlock:** Capped at 10,000 tokens; minting unlocks 21 days post-deployment;
- **100% On-Chain SVG Rendering:** Dynamic vector artwork and metadata computed purely on-chain with attributes `Starting multiplier` and `Status`.

5 Execution & Solvency Accounting

5.1 Dual-Liability Accounting System

RewardDistributor maintains two separate liability ledgers: Token Liabilities (`liability[asset]`): **D18{tok} dimension; Quote Liabilities (quoteLiability[asset]):** D18{quote} dimension.

Batch exchange rate:

$$\text{BatchRate} = \left\lfloor \frac{\text{TokensReceived} \cdot \text{RAY}}{\text{ClaimablePot}} \right\rfloor$$

Stalled swaps exceed timeout (`maxPendingAge`, 3 days default), enabling permissionless `fallbackFinalize()`.

5.2 On-Chain TWAP and Execution Bounds

1. **FATCAT Pair:** `PancakeV2TwapOracle` reads cumulative price for TWAP;
2. **bStock Pairs:** Time-weighted liquidity verification;
3. **Slippage Protection:** Router verifies:

$$\text{minOut} = \left\lfloor \frac{\text{expectedOut} \cdot (10000 - \text{deviationBps})}{10000} \right\rfloor$$

5.3 Conservative Rounding and Solvency Invariants

For each batch with $\sum_i x_i \leq S$:

$$\text{Rate} = \left\lfloor \frac{R \cdot \text{RAY}}{S} \right\rfloor, \quad r_i = \left\lfloor \frac{x_i \cdot \text{Rate}}{\text{RAY}} \right\rfloor$$

Guarantees:

$$\sum_i r_i \leq R$$

Tested property D1 invariants: `balanceOf(Distributor, a) ≥ liability[a]`

`balanceOf(Distributor, quote) ≥  $\sum_a$  quoteLiability[a]`

6 Security Posture & Dependencies

6.1 Restricted Vault Privileges

1. **The Belly:** Has immutable governor and guardian roles (governance 2-of-3 Gnosis Safe). Ops Safe receives 5/36 revenue atomically with zero protocol privileges. No admin sweep,

rescue, or withdrawal. Spender activation is one-time with timelock;

2. **StakingVault:** `redeem()` intentionally ignores pause. Pause blocks only new deposits.

6.2 Tax Processor External Ownership

Tax processor is retained by Flap factory post-graduation. Tax lasts for 100 years post-graduation (becoming permanently zero thereafter), and after a 1-year anti-farmer period taxes only the primary liquidity pool. Protocol provides `watch-processor.sh` to poll state. Direct custody risk affects future fees, not principal.

6.3 Default Diet and RWA Risk Isolation

1. **WBNB Default and Fallback:** Identity route requires no market swap, eliminating bStock RWA risk;
2. **bStock Credit Isolation:** 5% probation cap restricts exposure;
3. **FATCAT Listing:** Requires 3-day governance timelock post-graduation.

7 Verification & Disclaimers

7.1 Reproducible Test Evidence

Baseline: 77 test suites / 570 tests / 0 failures (FOUNDRY_PROFILE=local forge test). Property tests do not equate to formal verification.

7.2 Deterministic Scope and Non-Goals

1. **Not a Fixed-Income Product:** Zero APY commitments;
2. **Not a Lockup:** Principal withdrawable in any block;
3. **Zero Superlinear Bias:** Capital scales strictly linearly;
4. **Not a Pension:** Half-life 4.736 days under zero volume; 1.24% balance after 30 days.

8 Appendix A: Notation & Dimensions Index

Symbol	Semantic Definition	Domain Bound	/	EVM Dimension Scale
m	Global interval index	$\mathbb{N}^+$		Dimensionless (uint256)
j_i	Active from interval	$m_{\text{stake}} + 1$		Dimensionless (uint256)
p_i	Staked principal	$\geq 100,000$		D18{FAT-CAT}
$c_{i(m)}$	Seniority notch	Integer [1, 22]		Dimensionless scalar
$w_{i(m)}$	Effective weight	$p_i \cdot c_{i(m)}$		D18{weight}
$W(m)$	Total active weight	$\sum w_{i(m)}$		D18{weight}
W_{open}	Asset open denominator	$\sum W_{\text{open}} = W$		D18{weight}
P_{climb}	Climbing principal sum	$\sum p_i$		D18{FAT-CAT}

J_{climb}	Climbing weighted virtual index	$\sum(p_i \cdot \text{effectiveDiet}_i)$	D18{FAT-CAT}
P_{settled}	Settled principal sum	$\sum p_i$	D18{FAT-CAT}
headOpening	Opening interval head weight	$\sum(\text{head}_i \cdot p_i)$	D18{weight}
W_{exiting}	Residual exit weight	$\sum w_i$	D18{weight}
RAY	Accumulator scale	Constant 10^{27}	D27 scale
$\omega_{a,m}$	Yield per unit weight	Yield scalar	D27 (uint256)
$A_{a,m}$	Accumulator A	$\sum(1+k)\omega$	D27 prefix sum
$B_{a,m}$	Accumulator B	$\sum \omega$	D27 prefix sum
α	Belly release fraction	$\frac{8}{168} = \frac{1}{21}$	Dimensionless ($\approx 4.76\%$)
B_k	Belly available balance	Unreserved quote	D18{quote} (WBNB)
BatchRate	Batch conversion rate	Fixed exchange rate	D27 ratio
liability	Token liability	≥ 0	D18{tok}
quoteLiab	Quote asset liability	≥ 0	D18{quote}

9 Appendix B: Solidity Rounding and Solvency Bounds

9.1 Integer Floor Division

$$\forall a, b \in \mathbb{N}^+, \quad \left\lfloor \frac{a}{b} \right\rfloor = \frac{a - (a \bmod b)}{b} \leq \frac{a}{b}$$

Rounding error $\varepsilon = \frac{a \bmod b}{b}$ satisfies $0 \leq \varepsilon < 1$.

9.2 Non-negative Dust Theorem

Theorem: Given token purchase R , pot S , and $\sum_{i=1}^N x_i \leq S$:

$$\sum_{i=1}^N r_i \leq R$$

Proof:

1. Rate definition: $\text{Rate} = \left\lfloor \frac{R \cdot \text{RAY}}{S} \right\rfloor$
2. Position claim: $r_i = \left\lfloor \frac{x_i \cdot \text{Rate}}{\text{RAY}} \right\rfloor \leq \frac{x_i \cdot \text{Rate}}{\text{RAY}}$
3. Summing across N positions:

$$\sum_{i=1}^N r_i \leq \frac{\text{Rate}}{\text{RAY}} \sum_{i=1}^N x_i$$

4. Substituting $\sum x_i \leq S$ and $\text{Rate} \leq \frac{R \cdot \text{RAY}}{S}$:

$$\sum_{i=1}^N r_i \leq \frac{\text{Rate} \cdot S}{\text{RAY}} \leq \frac{\left(\frac{R \cdot \text{RAY}}{S}\right) \cdot S}{\text{RAY}} = R \quad \blacksquare$$

Corollary: Residual dust $\Delta_{\text{dust}} = R - \sum_{i=1}^N r_i \geq 0$ permanently settles in the distributor, ensuring physical balance dominates recorded liabilities ($\text{Balance} \geq \text{Liability}$).

9.3 Solvency Preconditions

1. **Standard Token Behavior:** No fee-on-transfer, no negative rebase;
2. **Liability Monotonicity:** `finalize()` and `fallbackFinalize()` increment atomically; `claim()` decrements atomically;
3. **Ownerless Distributor:** No administrative sweep or drain functions.

10 Appendix C: On-Chain Gas Profiling Benchmark

Operation / Call	Caller	Complexity	Measured Gas Units
<code>stake()</code> (initial)	User	$O(1)$	142,500
<code>stakeWithCertificate()</code>	Holder	$O(1)$	168,000
<code>stake()</code> (top-up)	User	$O(1)$	98,200
<code>redeem()</code> (full)	User	$O(1)$	125,600
<code>redeemAndIssueCertificate()</code>	User	$O(1)$	189,000
<code>setDiet()</code>	User	$O(1)$	68,400
<code>advance()</code>	Keeper	$O(M)$	185,000 (5 assets)
<code>claim()</code> (1 batch)	User	$O(1)$	88,300
<code>claim()</code> (10 batches)	User	$O(K)$	164,800
<code>claimThrough()</code>	User	$O(K_{\text{bd}})$	195,000 (bounded)
<code>claimNative()</code>	User	$O(K)$	112,000 (1 batch)

11 Appendix D: Verified Contracts Schedule

Component	Network	Verification Role
FATCAT Token	BNB Chain	1B fixed supply, immutable clone
PancakeSwap V2 Pool	BNB Chain	Locked liquidity, TWAP price feed
The Belly Vault	BNB Chain	Unprivileged exponential damped reservoir
StakingVault	BNB Chain	Isolated staking, unpausable redeem
SeniorityLedger	BNB Chain	$O(1)$ closed-form algebraic ledger
RewardAssetRegistry	BNB Chain	MENU registry with 5% probation cap
ExecutionRouter	BNB Chain	Delayed activation, MEV-protected router
RewardDistributor	BNB Chain	Dual-liability independent accounting

SeniorityCertificate (ERC-721)	BNB Chain	Genesis deployment; mint by transferring 100k FATCAT to DEAD on exit, 100% on-chain SVG
PancakeV2TwapOracle	BNB Chain	Endogenous TWAP price reader

12 Appendix E: Responsible Disclosure & Security Channel

1. **Official Security Channel:** Submit reports via GitHub Private Security Advisories;
2. **PGP Encryption:** Disclosures containing exploit proofs may be encrypted with the official PGP key;
3. **Review & Feedback:** Regular review with prompt triage feedback upon technical evaluation;
4. **White-Hat Grants:** Critical and High vulnerabilities receive discretionary grants disbursed from Ops Safe operational funds. Architecture is immutable; remediations follow governance migration procedures.

— Mathematics Belong to Everyone, Code is Law —